

SATYANSH GAUR

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EDUCATION

Dr. Akhilesh Das Gupta Institute of Technology (GGSIUP)

Expected 2028

B.Tech | CGPA: 8.6 / 10

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL

Backend & Systems: FastAPI, AsyncIO, multiprocessing, REST APIs, SQLite, TimescaleDB, PostgreSQL, PostGIS

ML / Data: NumPy, Numba, XGBoost, scikit-learn, PyTorch, pandas, xarray, dask, ChromaDB, NetworkX

Frontend & Tools: Next.js, React, Mapbox, Streamlit, Docker, Kubernetes, Git, Prometheus, Grafana

PROJECTS

Satellite Link Simulator

github.com/Satyanshgaur/Satellite-link-optimiser

- Engineered a high-throughput simulation engine processing 1,300+ real satellites from live data feeds; achieved 275,000 timesteps/sec through NumPy vectorisation — a 45× speedup over naive scalar loops (~6,000/sec).
- Identified a single bottleneck consuming 50.6% of runtime; rewrote it with Numba JIT compilation, reducing its share to 0.7% and yielding a 192× end-to-end speedup (96.1 ms → 0.5 ms).
- Designed a stateful handoff manager with configurable switching policies, hysteresis, and minimum dwell-time control, solving link-instability problems analogous to TCP connection management.
- Parallelised Monte Carlo workloads with ProcessPoolExecutor (~2.5× speedup); validated serial-vs-parallel output divergence at < 0.001%; ran 500,000-step simulations within 326 MB RAM.
- Integrated XGBoost link-quality prediction, regression test suite with deterministic seeds, interactive Streamlit dashboard, and Docker/Docker Compose deployment.

Orbit Ops- Neighborhood Air — NASA Space Apps 2025 (3rd Place)

github.com/Satyanshgaur/Satellite-link-optimiser

- Led a 5-member team to 3rd place globally at NASA Space Apps Hackathon 2025; built a hyperlocal air-quality forecasting platform fusing NASA TEMPO satellite data, EPA/OpenAQ sensors, and NOAA weather feeds.
- Architected a full-stack system: async FastAPI backend → ML forecasting pipeline (Random Forest / Gradient Boosting) → TimescaleDB + PostGIS time-series store → Next.js / Mapbox interactive map with personalised threshold alerts.
- Downscaled ~10 km satellite grids to street-level AQI predictions, enabling users to plan outdoor activity by time-of-day instead of relying on inaccurate citywide averages.

RecalNet — Persistent Graph Memory Framework (In Development)

github.com/Satyanshgaur/RecalNet

- Building a local-LLM-backed knowledge graph system: NetworkX + SQLite graph store, ChromaDB vector retrieval, and a hybrid retrieval layer combining semantic search with graph traversal (BFS up to 2 hops).
- Designed modular ingestion pipeline — entity/relation extraction → fuzzy node deduplication → confidence scoring → contradiction detection → FastAPI agent interface — using Qwen2.5-7B via Ollama on an RTX 3050 (6 GB VRAM).

OPEN SOURCE CONTRIBUTIONS

Arnio — C++ CSV Data-Preparation Engine

github.com/im-anishraj/arnio

- Contributed to a compiled C++ engine for parsing, type-inference, deduplication, and validation of messy CSVs; interoperates with pandas, NumPy, scikit-learn, DuckDB, and Arrow.

fenn — Friendly Environment for Neural Networks

github.com/pyfenn/fenn

- Contributed to a Python ML/DL automation framework; added prebuilt trainers, configuration management, and structured logging to reduce model-development boilerplate.

CERTIFICATIONS

- Columbia University (Columbia+ Online) — Machine Learning Certificate | Mathematics-focused: probabilistic models, optimisation, statistical learning theory
- Stanford University / Coursera — Machine Learning Specialisation (Andrew Ng) | Supervised learning, unsupervised learning, reinforcement learning

HONORS & AWARDS

- 3rd Place — NASA Space Apps Hackathon 2025 | Led 5-member team competing against 1,000+ global teams
- Finalist — Hyperspace Innovation Hackathon 2025
- Finalist — ISRO Bharatiya Antariksh Hackathon (BAH) 2025